Multi-Domain Multi-Task Rehearsal for Lifelong Learning

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Outline

- Background -

- Motivation -

- Methodology -

- Experiments -

Association for the Advancement of Artificial Intelligence

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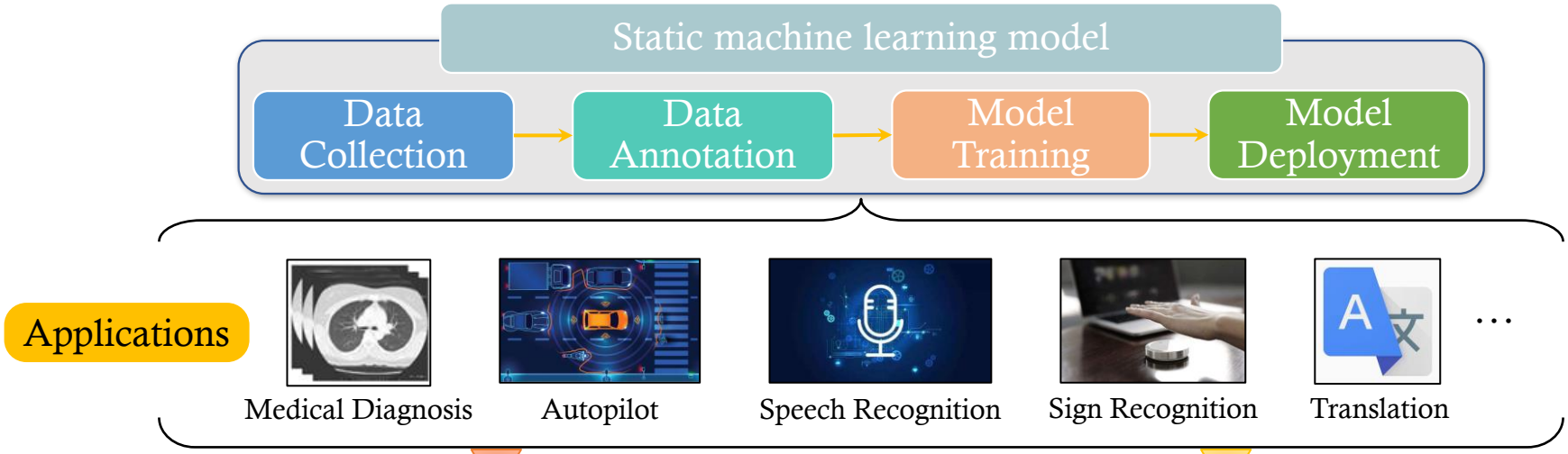
Static Machine Learning Model



Advantages

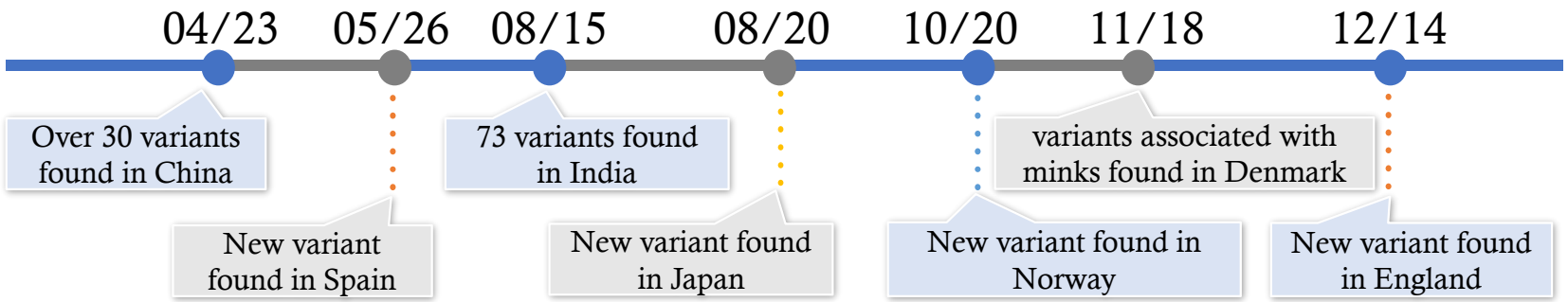
- Better Performance
- Easy Implementation
- Many Proven Algorithms

Requirements

- Large amount of data
- Manual annotations
- Recursive Training

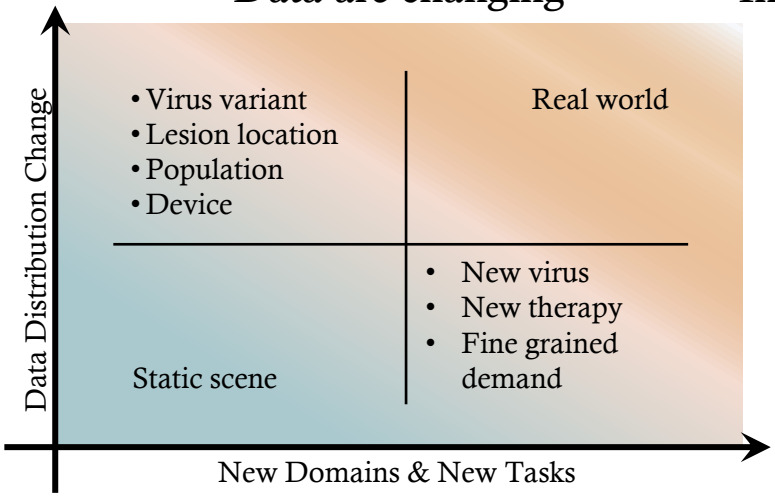
Can a static machine learning model satisfy the real world demand?

Data are Changing

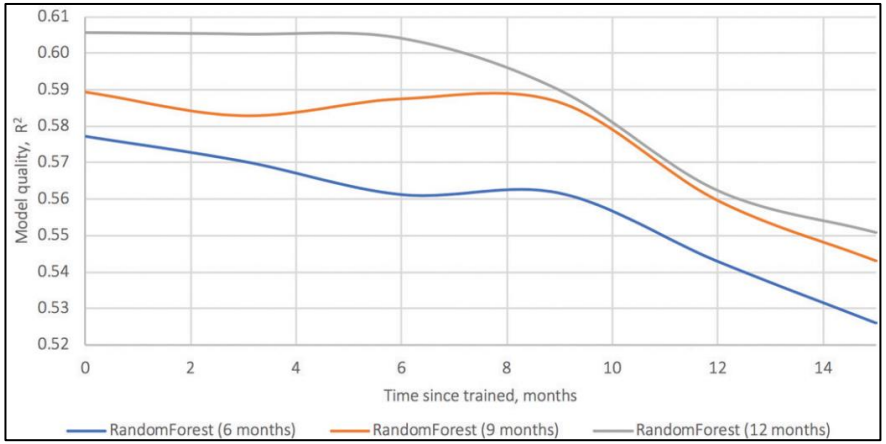


Multiple COVID-19 variants are circulating globally over time

Data are changing



In real scenarios, the model quality deteriorates over time*

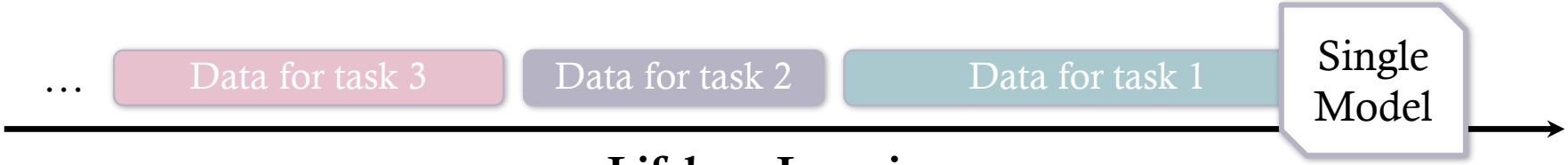


(3 Random Forest model predict the patient waiting time)

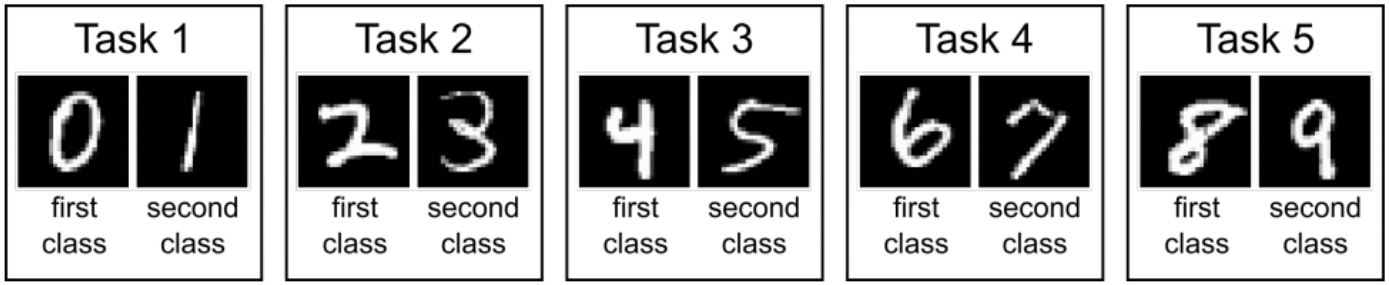
*Data Source: Department of Radiology, Massachusetts General Hospital, Harvard Medical School

Changed Data makes the static model unsustainable

Lifelong Learning

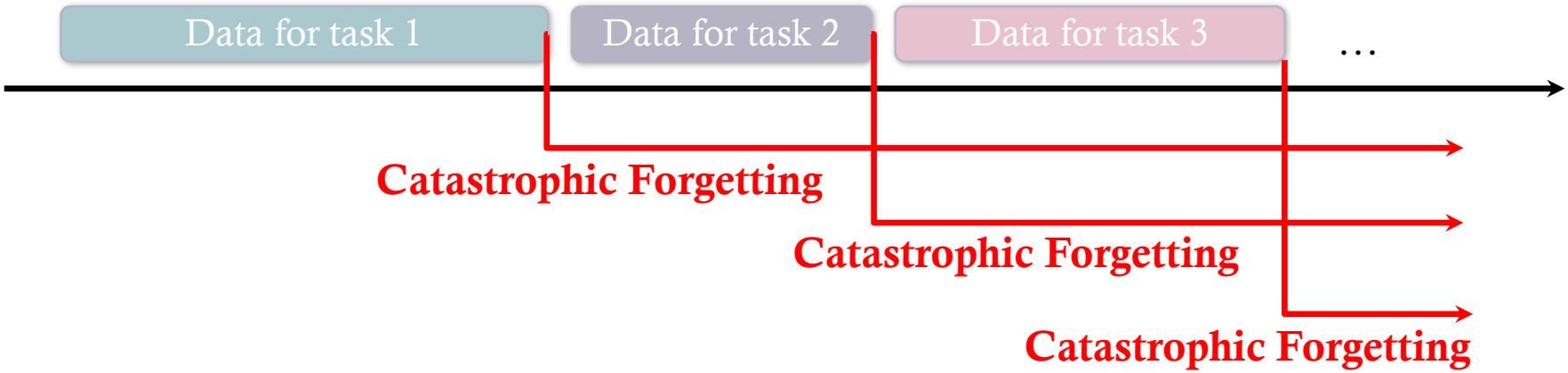


Lifelong Learning
(a.k.a. Continual Learning/Incremental Learning)
Learns new tasks/domains sequentially



Task-incremental	Solve tasks so far, task-ID provided	With task given, is it the 1st or 2nd class? (e.g., 0 or 1)
Domain-incremental	Solve tasks so far, task-ID not provided	With task unknown, is it a 1st or 2nd class? (e.g., in [0, 2, 4, 6, 8] or in [1, 3, 5, 7, 9])
Class-incremental	Solve tasks so far and infer task-ID	With task unknown, which digit is it? (i.e., choice from 0 to 9)

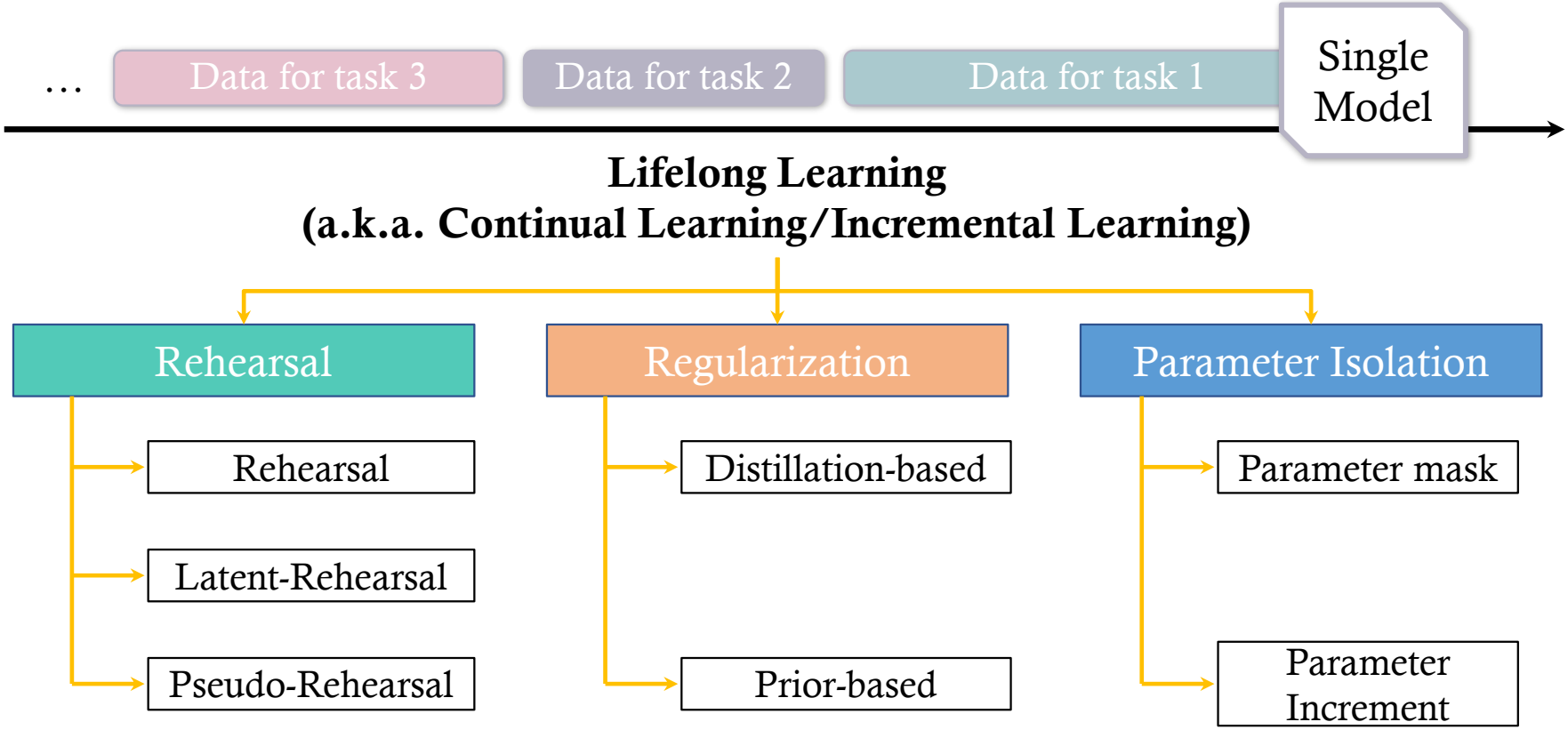
Catastrophic Forgetting



Catastrophic Forgetting

When trained on a new task, standard neural networks forget most of the information related to previously learned tasks.

Rehearsal



Goal of Lifelong Learning:
Mitigate the Catastrophic Forgetting

Previous Work: iCaRL

Algorithm 2 iCaRL INCREMENTALTRAIN

input $X^s, \ldots, X^t$ // training examples in per-class sets
input K // memory size
require Θ // current model parameters
require $\mathcal{P} = (P_1, \ldots, P_{s-1})$ // current exemplar sets

$\Theta \leftarrow$ UPDATEREPRESENTATION$(X^s, \ldots, X^t; \mathcal{P}, \Theta)$
$m \leftarrow K/t$ // number of exemplars per class
for $y = 1, \ldots, s-1$ **do**
  $P_y \leftarrow$ REDUCEEXEMPLARSET(P_y, m)
end for
for $y = s, \ldots, t$ **do**
  $P_y \leftarrow$ CONSTRUCTEXEMPLARSET(X_y, m, Θ)
end for
$\mathcal{P} \leftarrow (P_1, \ldots, P_t)$ // new exemplar sets

Save data into memory/update memory

Algorithm 3 iCaRL UPDATEREPRESENTATION

input $X^s, \ldots, X^t$ // training images of classes $s, \ldots, t$
require $\mathcal{P} = (P_1, \ldots, P_{s-1})$ // exemplar sets
require Θ // current model parameters
// form combined training set:

$$\mathcal{D} \leftarrow \bigcup_{y=s,\ldots,t} \{(x,y) : x \in X^y\} \cup \bigcup_{y=1,\ldots,s-1} \{(x,y) : x \in P^y\}$$

// store network outputs with pre-update parameters:
for $y = 1, \ldots, s-1$ **do**
  $q_i^y \leftarrow g_y(x_i)$ for all $(x_i, \cdot) \in \mathcal{D}$
end for
run network training (*e.g.* BackProp) with loss function

$$\ell(\Theta) = -\sum_{(x_i,y_i)\in\mathcal{D}} \Big[\sum_{y=s}^{t} \delta_{y=y_i} \log g_y(x_i) + \delta_{y\neq y_i} \log(1-g_y(x_i)) + \sum_{y=1}^{s-1} q_i^y \log g_y(x_i) + (1-q_i^y) \log(1-g_y(x_i)) \Big]$$

that consists of *classification* and *distillation* terms.

Rebuffi S A, Kolesnikov A, Sperl G, et al. iCaRL: Incremental classifier and representation learning[C]//Proceedings of the IEEE conference on Computer Vision and Pattern Recognition. 2017: 2001-2010.

Previous Rehearsal: GEM/AGEM

$$\arg\min_{\theta,\theta_t} \quad \ell(f_\theta, f_{\theta_t}, \mathcal{D}_t),$$
$$\text{s.t.} \quad \ell(f_\theta, f_{\theta_k}, \mathcal{M}_k) \leq \ell(f_\theta^{t-1}, f_{\theta_k}^{t-1}, \mathcal{M}_k), \quad \forall k < t$$

Transform

$$\text{minimize}_{\tilde{g}} \frac{1}{2} \quad \|g - \tilde{g}\|_2^2$$
$$\text{subject to} \quad \langle \tilde{g}, g_k \rangle \geq 0 \text{ for all } k < t.$$

Angle Constrains:

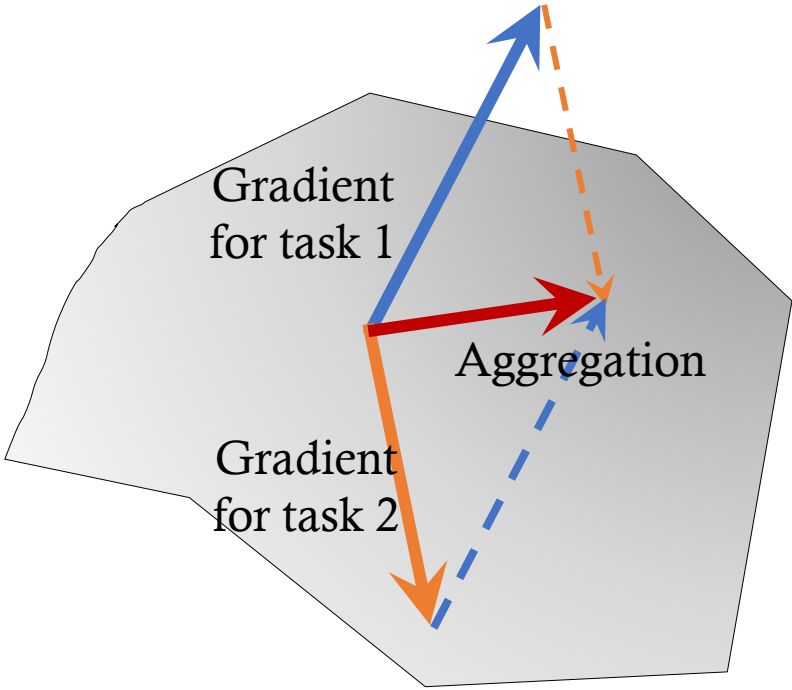
$$\langle g, g_k \rangle := \left\langle \frac{\partial \ell(f_\theta(x,t), y)}{\partial \theta}, \frac{\partial \ell(f_\theta, \mathcal{M}_k)}{\partial \theta} \right\rangle \geq 0, \text{ for all } k < t.$$

Lopez-Paz D, Ranzato M A. Gradient episodic memory for continual learning[C]//Advances in neural information processing systems. 2017: 6467-6476.

Previous Rehearsal: GEM/AGEM

GEM Quadratic Program (QP)

procedure TRAIN(f_θ, $\text{Continuum}_{\text{train}}$, $\text{Continuum}_{\text{test}}$)
  $\mathcal{M}_t \leftarrow \{\}$ for all $t = 1, \ldots, T$.
  $R \leftarrow 0 \in \mathbb{R}^{T \times T}$.
  for $t = 1, \ldots, T$ **do**:
    for (x, y) in $\text{Continuum}_{\text{train}}(t)$ **do**
      $\mathcal{M}_t \leftarrow \mathcal{M}_t \cup (x, y)$
      $g \leftarrow \nabla_\theta \, \ell(f_\theta(x, t), y)$
      $g_k \leftarrow \nabla_\theta \, \ell(f_\theta, \mathcal{M}_k)$ for all $k < t$
      $\tilde{g} \leftarrow$ PROJECT($g, g_1, \ldots, g_{t-1}$), see (11).
      $\theta \leftarrow \theta - \alpha \tilde{g}$.
    end for
    $R_{t,:} \leftarrow$ EVALUATE(f_θ, $\text{Continuum}_{\text{test}}$)
  end for
  return f_θ, R
end procedure



AGEM $\mathcal{M} = \cup_{k<t} \mathcal{M}_k$ → $\tilde{g} = g - \frac{g^\top g_{ref}}{g_{ref}^\top g_{ref}} g_{ref}$

Chaudhry A, Ranzato M A, Rohrbach M, et al. Efficient lifelong learning with a-gem[C]//ICLR, 2019.

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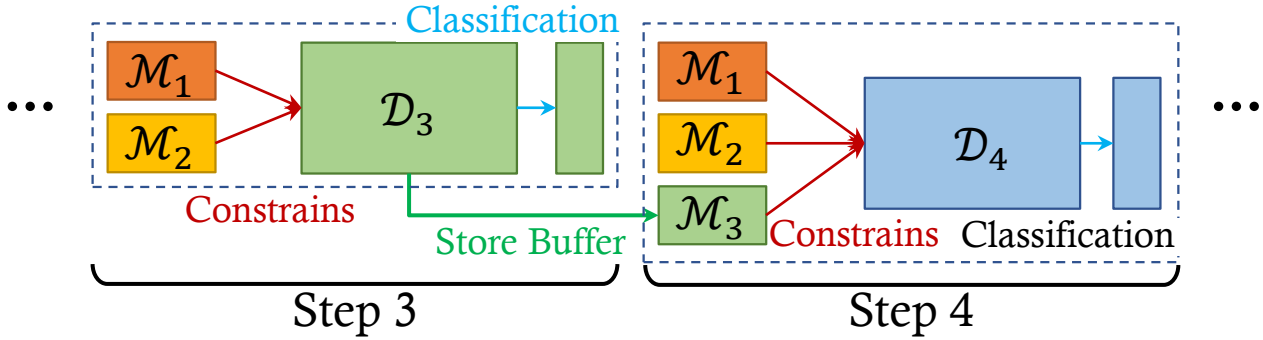
Association for the Advancement of Artificial Intelligence

Domain Shift in Feature Space



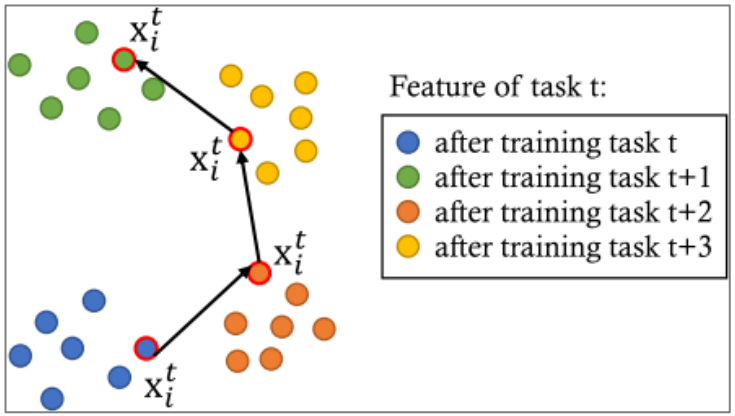
Rehearsal in Lifelong Learning

Domain shift in feature space:

- Data imbalance
- Task isolation

Result in:

task indistinguishable



Domain shift in feature space

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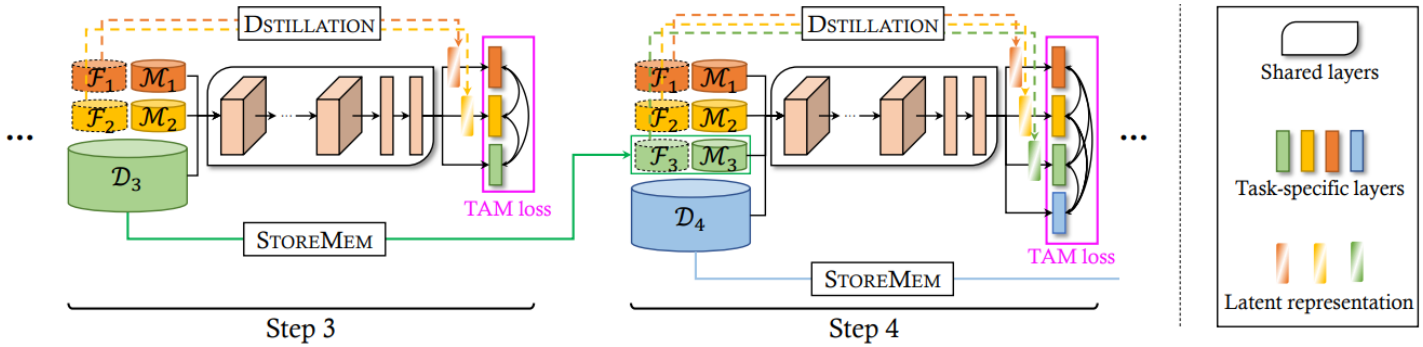
Pipeline



1. Multi-domain Multi-task Rehearsal
2. [TAM] Two-level angular margin loss
3. [DISTILLATION] Episodic distillation (Optional)

MDMTR

GEM defined:

$$\arg\min_{\theta,\theta_t} \quad \ell(f_\theta, f_{\theta_t}, \mathcal{D}_t),$$

$$\text{s.t.} \quad \ell(f_\theta, f_{\theta_k}, \mathcal{M}_k) \leq \ell(f_\theta^{t-1}, f_{\theta_k}^{t-1}, \mathcal{M}_k), \quad \forall k < t$$

We defined:

$$\arg_{\theta,\{\theta_1,\cdots,\theta_t\}} \quad \{\min \ell(f_\theta, f_{\theta_t}, \mathcal{D}_t), \min \ell(f_\theta, f_{\theta_k}, \mathcal{M}_k), \cdots,$$

$$\min \ell(f_\theta, f_{\theta_1}, \mathcal{M}_1)\},$$

$$\text{s.t.} \quad d(f_i, f_j) \geq d(f_i^{t-1}, f_j^{t-1}), i, j \in [1, t], i \neq j,$$

Notations

$$f_i = f_\theta(\mathcal{D}_i) \text{ if } i = t$$

$$f_i = f_\theta(\mathcal{M}_i) \text{ if } i < t.$$

Cross Domain Softmax

Multi-objective to Single-objective optimization:

$$\arg\min_{\theta} \quad \ell(f_\theta, f_{\theta_t}, \mathcal{D}_t) + \sum_{k=1}^{t-1} \ell(f_\theta, f_{\theta_k}, \mathcal{M}_k)$$

Cross Domain Softmax:

$$\ell_k = -\frac{1}{N_k} \sum_{n=1}^{N_k} \log \frac{e^{\left(W_{y_n}^k\right)^T x_n + b_{y_n}}}{\sigma_n},$$

where

$$\sigma_n = \boxed{\sum_{j=1}^{C_k} e^{\left(W_j^k\right)^T x_n + b_j}} + \boxed{\sum_{i=1, i\neq k}^{t} \sum_{j=1}^{C_i} e^{\left(W_j^i\right)^T x_n + b_j}}$$

Native softmax term Cross domain term

Two-level Angular Margin (TAM) Loss

Arcface:

$$\ell = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{e^{s\cdot\cos\left(\theta_{y_i}+m\right)}}{e^{s\cdot\cos\left(\theta_{y_i}+m\right)}+\sum_{j=1,j\neq y_i}^{n}e^{s\cdot\cos\theta_j}}$$

Ours:

$$\ell_k = -\frac{1}{N_k}\sum_{n=1}^{N_k}\log\frac{e^{s\cdot\cos\left((\theta_{y_n}^k+m^{\mathrm{c}})+m^{\mathrm{t}}\right)}}{\sigma_n},$$

where

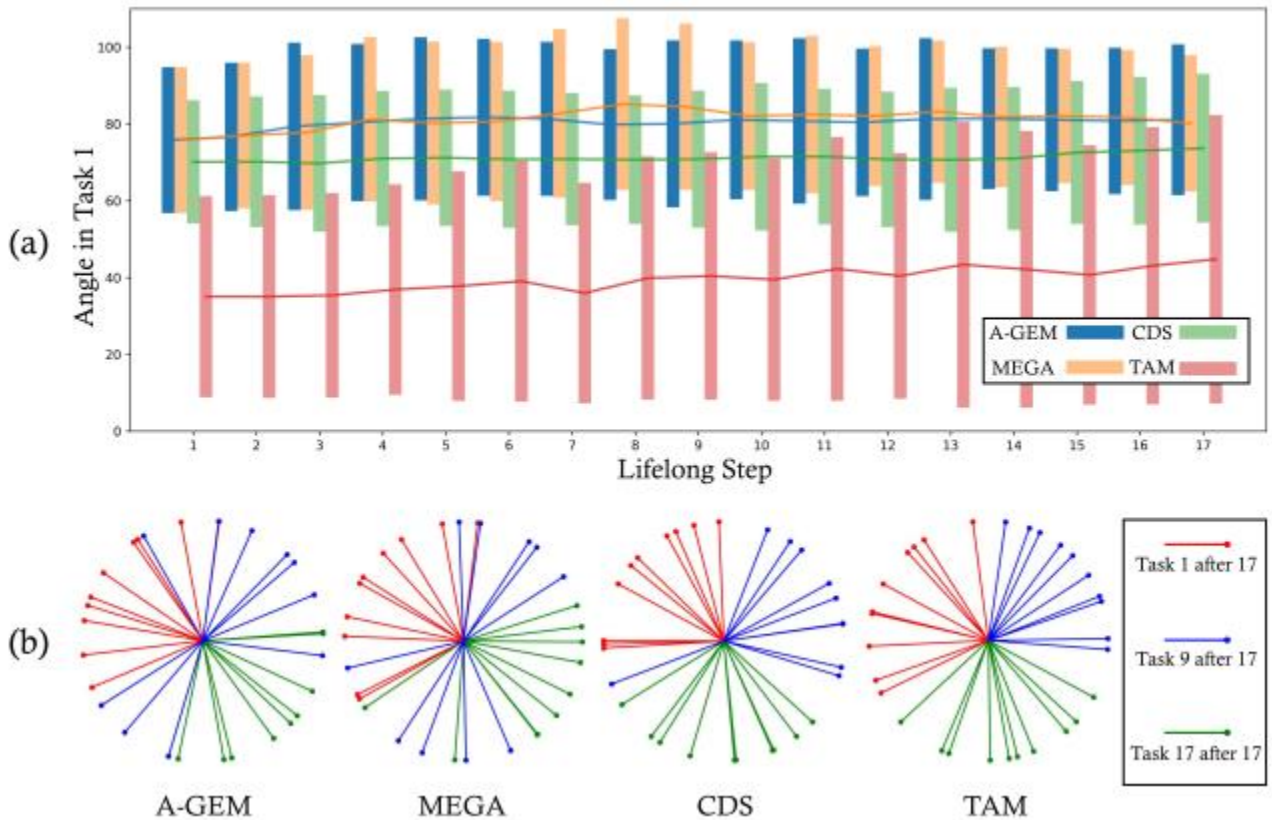
$$\sigma_n = e^{s\cdot\cos\left((\theta_{y_i}^k+m^{\mathrm{c}})+m^{\mathrm{t}}\right)}+\sum_{j=1,j\neq y_i}^{C_k}e^{s\cdot\cos(\theta_j^k+m^{\mathrm{t}})}+\sum_{i=1,i\neq k}^{t}\sum_{j=1}^{C_i}e^{s\cdot\cos\theta_j^i}.$$

Two level margins

Class level: m_c

Task level: m_t

Two-level Angular Margin (TAM) Loss



On Permuted MNIST, (a) the changes of angle range between feature and the target weight center of task 1 along the lifelong learning; (b) the angulars relations of class centers of task 1, 9 and 17 after trained on task 17

(Optional) Episodic Distillation

Episodic Distillation loss

$$\arg\min_{\theta} \quad \ell(f_\theta, f_{\theta_t}, \mathcal{D}_t) + \sum_{k=1}^{t-1} \left[\ell(f_\theta, f_{\theta_k}, \mathcal{M}_k) + \boxed{\tilde{\ell}(f_\theta, \mathcal{F}_k)} \right] \tag{9}$$

where

$$\tilde{\ell}(f_\theta, \mathcal{F}_k) \triangleq \frac{1}{N_k} \sum_i \tilde{\ell}_i(f_\theta(\mathbf{x}_{k,i}), \mathbf{f}_{k,i}). \tag{10}$$

Algorithm

Procedure TRAIN(f_θ, $f_{\theta_{1:T}}$, $\{\mathcal{D}_1^{\text{trn}}, \cdots, \mathcal{D}_T^{\text{trn}}\}$)
  $\mathcal{M}, \mathcal{F} \leftarrow \{\}$
  for $t = 1$ **to** T **do**
    for $(\mathbf{x}, y) \in \mathcal{D}_t^{\text{trn}}$ **do**
      $g, g_1 \leftarrow \nabla_\theta \ell(f_\theta(\mathbf{x}, t), y)$
      if $t = 1$ **then**
        $\tilde{g} \leftarrow g$
      else
        $g^{\text{ref}}, g_{1:t-1} \leftarrow \nabla_\theta \ell(f_\theta, f_{\theta_{1:t-1}}, \mathcal{M})$
        $g^{\text{ref}} \leftarrow g^{\text{ref}} + \nabla_\theta \tilde{\ell}(f_\theta, \mathcal{F}_{\text{ref}})$
        $\tilde{g} \leftarrow g + g^{\text{ref}}$
      end if
      $\theta \leftarrow \theta - \text{StepSize} \cdot \tilde{g}$
      $\theta_{1:t} \leftarrow \theta_{1:t} - \text{StepSize} \cdot g_{1:t}$
    end for
    $\mathcal{M}, \mathcal{F} \leftarrow$ STOREMEM($\mathcal{M}, \mathcal{F}, \mathcal{D}_t^{\text{trn}}, f_\theta$)
  end for

Procedure STOREMEM($\mathcal{M}, \mathcal{F}, \mathcal{D}, f$)
  for $i = 1$ **to** $|\mathcal{M}|/T$ **do**
    $(\mathbf{x}, y) \sim \mathcal{D}$
    $\mathcal{M} \leftarrow \mathcal{M} + (\mathbf{x}, y)$
    $\mathcal{F} \leftarrow \mathcal{F} + f(\mathbf{x})$
  end for
  Return $\mathcal{M}, \mathcal{F}$

Procedure EVAL(f_θ, $f_{\theta_{1:T}}$, $\{\mathcal{D}_1^{\text{tst}}, \cdots, \mathcal{D}_T^{\text{tst}}\}$)
  $a \leftarrow 0 \in \mathbb{R}^T$
  for $t = 1$ **to** T **do**
    $a_t \leftarrow 0$
    for $(\mathbf{x}, y) \in \mathcal{D}_t^{\text{tst}}$ **do**
      $a_t \leftarrow a_t + \text{Accuracy}(f_{\theta_t}(f_\theta(\mathbf{x}, t)), y)$
    end for
    $a_t \leftarrow a_t / |\mathcal{D}_t^{\text{tst}}|$
  end for
  Return a

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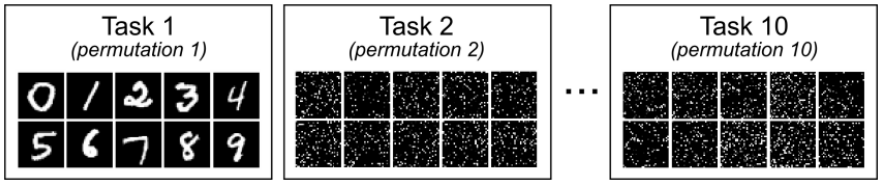
Datasets

Permuted MNIST

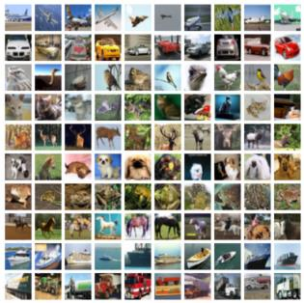


Split CUB

CUB200-2011
(200 classes)

Split into
10 classes × 20 tasks

Split CIFAR



CIFAR 100

Split into
5 classes × 20 tasks

Split AWA

AWA
(50 classes)

Split into
5 classes × 20 tasks
(with replacement)

Metrics

Average Accuracy: Average accuracy after the model has been trained continually with all the mini-batches up till task t ∈ {1, …, T} is defined as

$$A_t = \frac{1}{t}\sum_{j=1}^{t} a_{t,B_t,j}$$

Forgetting Measure: Average forgetting after the model has been trained continually with all the mini-batches up till task t ∈{1, …, T} is defined as

$$F_t = \frac{1}{t-1}\sum_{j=1}^{t-1} f_j^t$$

$$f_j^t = \max_{l\in\{1,\cdots,k-1\}} a_{l,B_l,j} - a_{t,B_k,j}$$

Learning Curve Area: The area of the convergence curve for any average b-shot performance after the model has been trained for all the T tasks, where b ∈ [0, β].

$$\mathrm{LCA}_\beta = \frac{1}{\beta+1}\int_0^\beta Z_b db = \frac{1}{\beta+1}\sum_{b=0}^{\beta} Z_b,$$

$$Z_b = \frac{1}{T}\sum_{t=1}^{T} a_{t,b,t}$$

Long-Term Remembering: LTR quantifies the accuracy drop on each task relative to the accuracy just right after the task has been learned

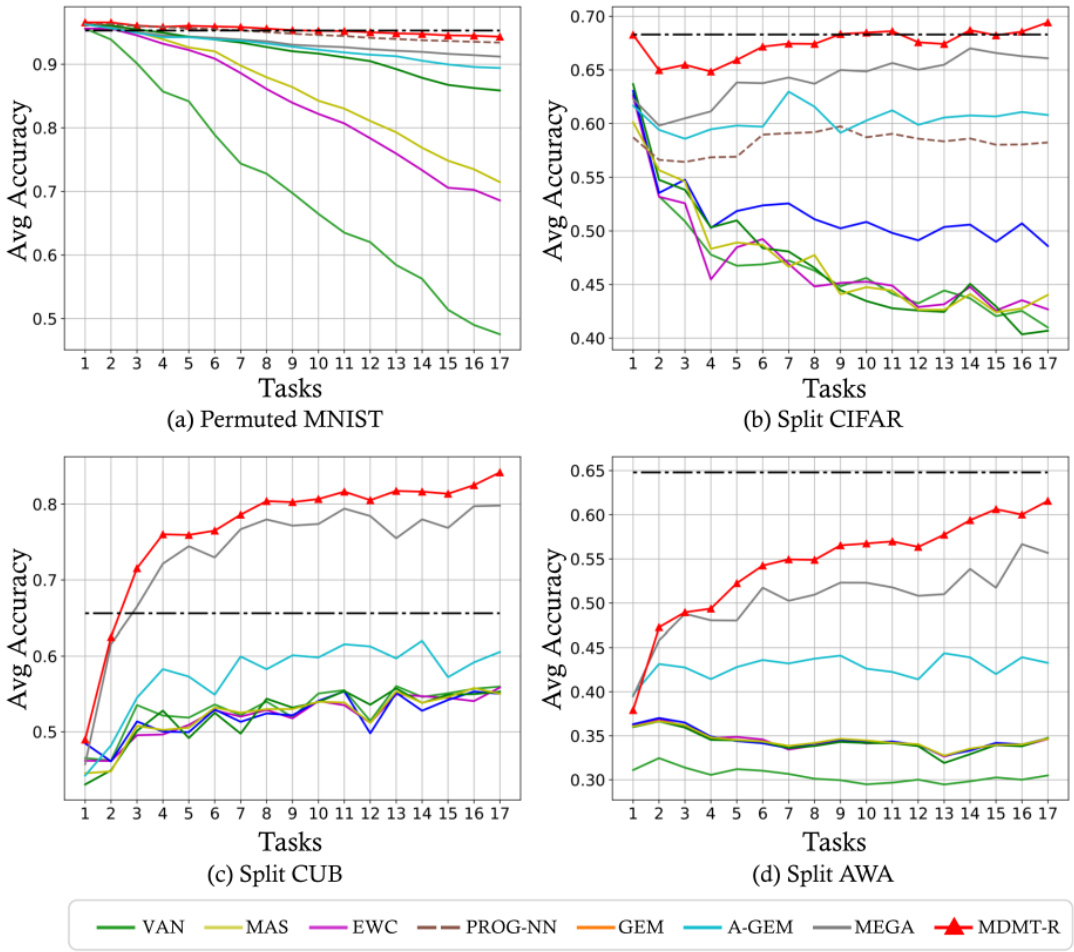
$$\mathrm{LTR} = -\frac{1}{T-1}\sum_{j=1}^{T-1}(T-j)\min\left\{0, a_{T,B_T,j} - a_{j,B_j,j}\right\}$$

Main Results

Method	Permuted MNIST				Split CIFAR			
	$A_T(\%)$	F_T	LCA_{10}	LTR	$A_T(\%)$	F_T	LCA_{10}	LTR
Joint	95.30	-	-	-	68.30	-	-	-
VAN	47.55 ± 2.37	0.52 ± 0.026	0.259 ± 0.005	5.375 ± 0.194	40.44 ± 1.02	0.27 ± 0.006	0.309 ± 0.011	2.613 ± 0.174
EWC	68.68 ± 0.98	0.28 ± 0.010	0.276 ± 0.002	3.292 ± 0.135	42.67 ± 4.24	0.26 ± 0.039	0.336 ± 0.010	2.493 ± 0.427
MAS	70.30 ± 1.67	0.26 ± 0.018	0.298 ± 0.006	-	42.35 ± 3.52	0.26 ± 0.030	0.332 ± 0.010	-
RWalk	85.60 ± 0.71	0.08 ± 0.007	$\mathbf{0.319} \pm 0.003$	-	42.11 ± 3.69	0.27 ± 0.032	0.334 ± 0.012	-
MER	-	-	-	-	37.27 ± 1.68	$\mathbf{0.03} \pm 0.030$	0.051 ± 0.101	-
GEM	89.50 ± 0.48	0.06 ± 0.004	0.230 ± 0.005	-	61.20 ± 0.78	0.06 ± 0.007	0.360 ± 0.007	-
A-GEM	89.32 ± 0.46	0.07 ± 0.004	0.277 ± 0.008	0.716 ± 0.048	61.28 ± 1.88	0.09 ± 0.018	0.350 ± 0.013	0.643 ± 0.124
ER	90.47 ± 0.14	0.03 ± 0.001	0.184 ± 0.004	0.367 ± 0.013	63.97 ± 1.30	0.06 ± 0.006	0.349 ± 0.105	0.451 ± 0.333
MEGA	91.21 ± 0.10	0.05 ± 0.001	0.283 ± 0.004	0.524 ± 0.017	66.12 ± 1.94	0.06 ± 0.015	$\mathbf{0.375} \pm 0.012$	0.356 ± 0.114
MDMT-R	$\mathbf{94.33} \pm 0.04$	$\mathbf{0.02} \pm 0.000$	0.298 ± 0.003	$\mathbf{0.247} \pm 0.009$	$\mathbf{69.20} \pm 1.60$	0.04 ± 0.010	0.334 ± 0.008	$\mathbf{0.283} \pm 0.099$

Method	Split CUB				Split AWA			
	$A_T(\%)$	F_T	LCA_{10}	LTR	$A_T(\%)$	F_T	LCA_{10}	LTR
Joint	65.60	-	-	-	64.80	-	-	-
VAN	53.89 ± 2.00	0.13 ± 0.020	0.292 ± 0.008	0.976 ± 0.215	30.35 ± 2.81	0.04 ± 0.013	0.214 ± 0.008	0.202 ± 0.090
EWC	53.56 ± 1.67	0.14 ± 0.024	0.292 ± 0.009	1.021 ± 0.210	33.43 ± 3.07	0.08 ± 0.021	0.257 ± 0.011	0.675 ± 0.214
MAS	54.12 ± 1.72	0.13 ± 0.013	0.293 ± 0.008	-	33.83 ± 2.99	0.08 ± 0.022	0.257 ± 0.011	-
RWalk	54.11 ± 1.71	0.13 ± 0.013	0.293 ± 0.009	-	33.63 ± 2.64	0.08 ± 0.023	0.258 ± 0.011	-
PI	55.04 ± 3.05	0.12 ± 0.026	0.292 ± 0.010	-	33.86 ± 2.77	0.08 ± 0.022	0.259 ± 0.011	-
A-GEM	61.82 ± 3.72	0.08 ± 0.021	0.302 ± 0.011	0.456 ± 0.174	44.95 ± 2.97	0.05 ± 0.014	0.287 ± 0.012	0.178 ± 0.082
ER	73.63 ± 0.52	$\mathbf{0.01} \pm 0.005$	0.265 ± 0.004	$\mathbf{0.001} \pm 0.001$	54.27 ± 4.05	$\mathbf{0.02} \pm 0.030$	0.293 ± 0.009	0.014 ± 0.015
MEGA	80.58 ± 1.94	$\mathbf{0.01} \pm 0.017$	0.311 ± 0.010	0.002 ± 0.002	54.28 ± 4.84	0.05 ± 0.040	$\mathbf{0.305} \pm 0.015$	0.070 ± 0.114
MDMT-R	$\mathbf{84.27} \pm 1.63$	$\mathbf{0.01} \pm 0.015$	$\mathbf{0.337} \pm 0.013$	0.017 ± 0.014	$\mathbf{61.56} \pm 3.36$	$\mathbf{0.02} \pm 0.027$	0.298 ± 0.008	$\mathbf{0.002} \pm 0.002$

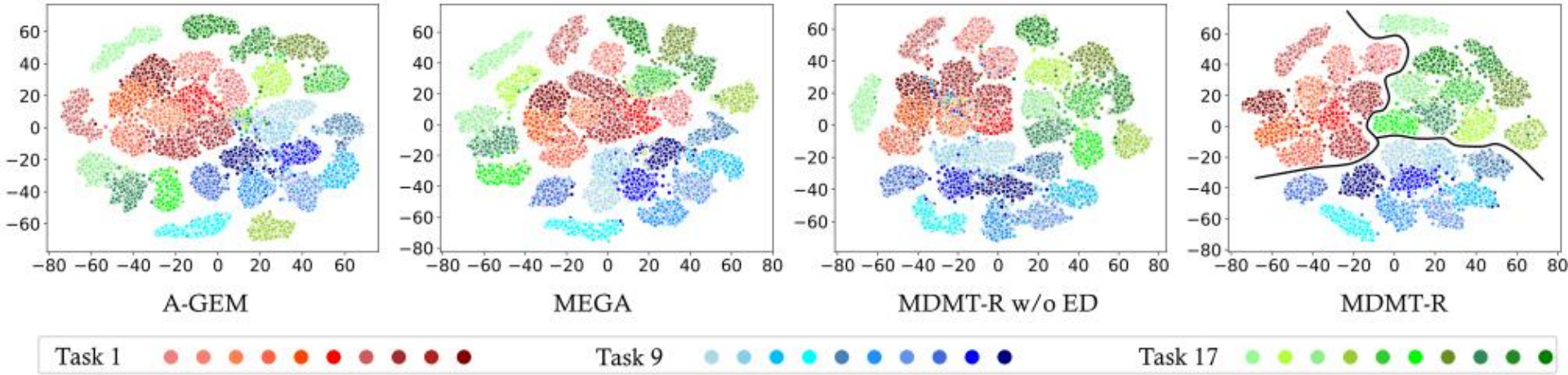
Main Results



Ablation Studies

m_t	m_c	ED	$A_T(\%)$	F_T	LCA_{10}	LTR
-	-	-	65.44 ± 1.13	0.052 ± 0.006	$\mathbf{0.371} \pm 0.008$	0.377 ± 0.076
-	-	✓	66.44 ± 2.22	0.050 ± 0.009	0.370 ± 0.014	0.307 ± 0.066
0.0	0.0	-	67.15 ± 2.02	0.053 ± 0.012	0.353 ± 0.006	0.411 ± 0.097
0.1	0.0	-	67.49 ± 1.55	$\mathbf{0.049} \pm 0.010$	0.354 ± 0.005	0.369 ± 0.096
0.0	0.01	-	67.45 ± 1.09	0.059 ± 0.008	0.354 ± 0.005	0.483 ± 0.064
0.1	0.01	-	67.68 ± 1.72	0.052 ± 0.008	0.350 ± 0.007	0.390 ± 0.062
0.4	0.01	-	67.28 ± 0.97	0.053 ± 0.012	0.347 ± 0.006	0.394 ± 0.106
0.4	0.05	-	66.68 ± 1.23	0.063 ± 0.005	0.333 ± 0.005	0.473 ± 0.077
0.4	0.1	-	64.97 ± 1.13	0.084 ± 0.009	0.324 ± 0.008	0.680 ± 0.086
0.1	0.01	✓	$\mathbf{68.64} \pm 1.35$	0.059 ± 0.016	0.334 ± 0.008	$\mathbf{0.297} \pm 0.103$

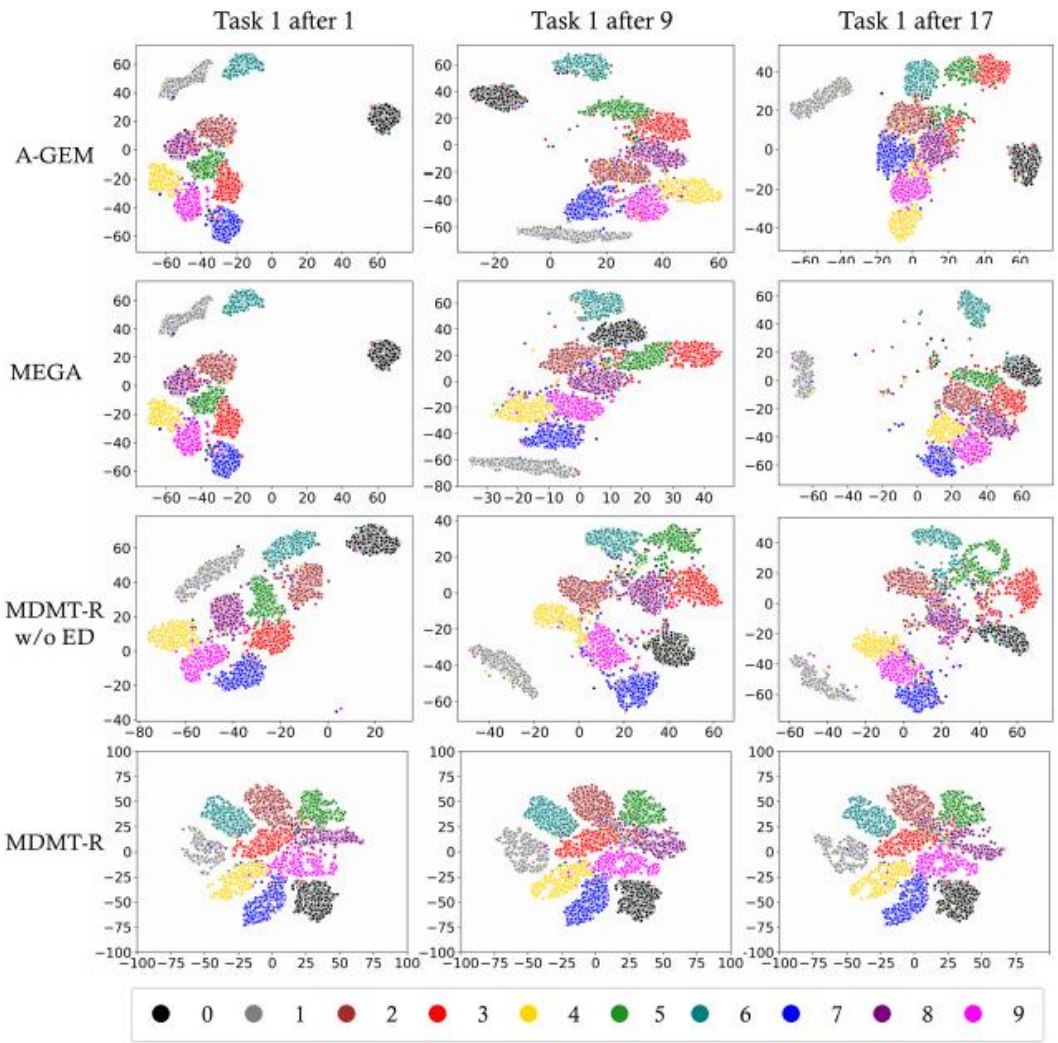
Visualization



Final t-SNE of the features extracted from task 1, 9 and 17 on Permuted MNIST after the training on task 17.

Visualization

t-SNE of the features from task 1 on Permuted MNIST after the lifelong learning on task 1, 9 and 17.



Thanks!

Association for the
Advancement of Artificial Intelligence

Computational Cost and Memory Complexity

Method	Inference time (ms)				Memory	
	MNIST	CIFAR	CUB	AWA	Training	Testing
VAN	-	-	-	-	$P + B * H$	$P + B * H$
EWC	-	-	-	-	$4 * P + B * H$	$4 * P + B * H$
GEM	-	-	-	-	$P * T + (B + M) * H$	$P + B * H$
A-GEM	32.58	28.74	81.83	97.47	$2 * P + (B + M) * H$	$P + B * H$
MEGA	37.63	26.38	84.96	109.67	$2 * P + (B + M) * H$	$P + B * H$
MDMT-R	35.16	30.09	90.47	97.51	$2 * P + (B + M) * H + M * H_l$	$P + B * H$